

Separating arrival rounding, source startup, and reverberation in three-microphone azimuth estimation

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Abstract

Purpose. Small-array simulations can mix propagation discretization, source history, and acoustic effects. We quantify these contributions for a fixed three-microphone azimuth estimator. **Methods.** Independent analytical and image-source references verify a fractional-delay shoebox model. A frozen campaign contains 276,480 GCC-PHAT frame evaluations, paired rounding/startup/reflection controls, and independent records for averaging. Uncertainty resamples whole recordings within fixed scenes. **Results.** Confirmatory numerical refinements change checked estimates by less than 0.021° . At 10 dB direct-referenced SNR, direct-path RMSE increases from 0.929° to 2.279° after arrival rounding. In the high-reverberation subset, first-frame RMSE is 4.59° at source onset versus 40.42° with populated prehistory. Averaging 128 frames reduces held-out RMSE from 1.60° to 0.84° in the low subset and from 39.76° to 10.04° in the high subset. A geometry-only timing bound accounts for shared-microphone dependence. **Conclusion.** Source history can create a large apparent accuracy benefit; arrival rounding introduces a separate, measurable distortion. Residual errors depend on the chosen acoustic scenes and estimator. These computations establish neither a universal reverberation floor nor hardware performance. **Keywords:** acoustic

localization; GCC-PHAT; image-source method; numerical verification; source history; uncertainty

1 Introduction

Estimating source direction from a compact microphone array appears to be a simple delay problem. The simulation used to evaluate it is less simple: it must construct microphone signals, represent fractional propagation times, choose an observation interval, estimate delays, and interpret them through an array model. An error introduced at any one of these steps may be attributed incorrectly to room acoustics. This is especially consequential when the acoustic travel time across the array spans only a small number of samples.

Generalized correlation is an established approach to time-delay estimation [1]. Equilateral three-microphone arrays combined with GCC-PHAT have already been studied experimentally and numerically [2]. Fractional-delay filtering is also established [3]. The present work proposes neither a new estimator nor a new microphone geometry. Its purpose is to make a particular computational evaluation traceable and to test which conclusions survive numerical and recording-protocol controls.

The image-source method provides an explicit model of specular shoebox propagation [4]. It allows matched experiments that would be difficult to realize physically, such as retaining the same source and noise while removing every reflection. However, an image model still requires reflection coverage, a duration cutoff, and a representation of noninteger arrival times. A simulator can also change the problem by beginning each analysis frame with an empty room. Numerical verification

and physical validation therefore answer different questions. Recent acoustic benchmarking emphasizes the distinction between implementation errors, numerical convergence, and differences in physical assumptions [5].

We study three specific choices: rounding each path arrival to the sample grid, omitting source prehistory, and including reflections. Exact plane-wave and spherical controls separate algebraic recovery from finite-distance curvature. Independent room enumeration and refinement tests establish numerical resolution before a confirmatory campaign. Paired records identify changes caused by the three choices, while separate long recordings test averaging. A deterministic precision criterion relates bounded microphone-arrival errors to geometry without assuming independent pair-delay errors.

The contribution is a verified attribution and replication study with inspectable frame results, not an exhaustive survey or a claim of priority. The scope is one stationary synthetic source, ideal matched microphones, horizontal azimuth, and three prescribed room geometries. Real recordings, moving or multiple sources, hardware timing, microphone mismatch, and scattering are outside the evaluation.

2 Physical and signal-processing model

2.1 Array and arrival geometry

The three microphone positions relative to the array center are

$$\mathbf{m}_i = R[\cos \phi_i, \sin \phi_i]^T, \quad \phi_i \in \{90^\circ, 210^\circ, 330^\circ\}, \quad (1)$$

with circumradius $R = 0.05$ m and pair spacing $\sqrt{3}R = 0.0866$ m. Sound speed is fixed at $c = 343$ m/s. The source is 1.5 m from the array center, at the microphone height. The direct spherical arrival is $t_i = \|\mathbf{s} - \mathbf{m}_i\|/c$, using horizontal relative coordinates. For an ideal plane wave with unit source direction $\mathbf{u}$, $t_i = -\mathbf{m}_i^T \mathbf{u}/c$ up to an irrelevant common delay.

For pairs $i < j$, define $\tau_{ij} = t_i - t_j$ and let the rows of A be $(\mathbf{m}_i - \mathbf{m}_j)^T$. The estimator interprets its three pair delays through

$$\tilde{\mathbf{u}} = -cA^+ \hat{\boldsymbol{\tau}}, \quad \hat{\theta} = \text{atan2}(\tilde{u}_y, \tilde{u}_x). \quad (2)$$

All pairs have equal weight. The direction vector is not constrained to unit norm before taking its angle. Thus, exact spherical arrivals can retain a finite-distance mismatch when interpreted by this planar far-field solve. This contribution is not an interpolation error.

2.2 Room responses

Each room is a three-dimensional rectangular enclosure with six frequency-independent pressure-reflection coefficients. For room side lengths L_a , source coordinates s_a , integers $n_a \in [-K, K]$, and parities $p_a \in \{0, 1\}$, the image coordinate is

$$s_a^{\text{img}} = 2n_a L_a + (1 - 2p_a)s_a. \quad (3)$$

The corresponding low/high wall reflection counts are $|n_a - p_a|$ and $|n_a|$. A path of length r has delay r/c and pressure gain $\prod_w \beta_w^{N_w}/(4\pi r)$. Paths longer than cT_h are omitted. The main settings are $K = 48$ and $T_h = 0.9$ s. These choices are based on refinement, rather than a nominal reflection order alone.

Uniform wall absorption uses the nominal Sabine relation $\alpha = 0.161V/(ST_{\text{nom}})$ and $\beta = \sqrt{1 - \alpha}$, where V and S are volume and surface area. T_{nom} labels an absorption setting; it is not asserted to equal the realized reverberation time. Calibration of such relationships is itself a room-model problem [6]. Frequency-independent specular reflection is a deliberate simplification: diffraction, diffuse scattering, air attenuation, microphone directivity, and material phase responses are absent.

For fractional arrivals, each path is deposited linearly on a time grid 16 times finer than the acquisition grid. A normalized Kaiser-windowed sinc decimation filter, with shape parameter 8.6 and half-width 32 acquisition samples, then produces the response. A separate rounded condition deposits each path at $\text{round}(f_s r/c)$ without that reconstruction. The same image coverage and gains are used. An explicit switch produces direct-only responses, which retain the same array/source geometry and overall response length.

2.3 Continuous source and noise

Sampling is at 48 kHz. The source is a Gaussian sequence projected by an FFT mask onto 300–3400 Hz and normalized in RMS. Each record includes 45,282 samples of source prehistory before the analysis interval. Convolution uses the full response. All microphone signals then pass through the same causal fourth-order Butterworth bandpass at 300–3400 Hz. The noise channels originate from independent Gaussian sequences filtered through the same acquisition response.

Noise power is calibrated using the acquired, steady, direct-only signal averaged across the three microphones and the full analysis interval. The specified SNR is the ratio of this reference power to the acquired noise power. Adding reflections therefore does not change the noise realization or its scaling within a paired comparison. This SNR should not be interpreted as the ratio of the complete reverberant signal power to noise power.

Steady records retain the source prehistory. Onset records use exactly the same source samples inside the analysis interval but zero the preceding source samples once. Both retain the same noise. Switching the source also introduces an onset spectrum; the experiment includes that physical transient. The primary analysis uses contiguous, nonoverlapping 2048-sample frames, each 42.667 ms long. A separately labeled exploratory diagnostic resets the source and acquisition history for every frame; it is not treated as a continuous recording.

2.4 GCC-PHAT implementation

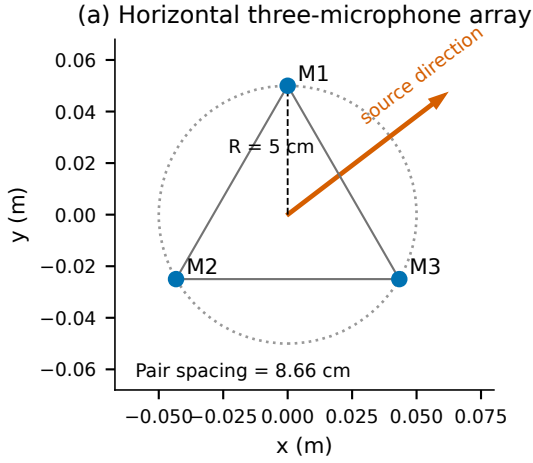
Each microphone frame is mean-centered and multiplied by a symmetric Hann window. The transform size is the next FFT-efficient size at least twice the frame length, here 4096. With $G_{ij}[k] = X_i[k]X_j^*[k]$, the regularized cross spectrum is

$$W_{ij}[k] = \frac{G_{ij}[k]}{\max\{|G_{ij}[k]|, 10^{-3} \max_l |G_{ij}[l]| + 10^{-12}\}}, \quad (4)$$

and is zeroed outside 300–3400 Hz. The regularization maximum is computed before band exclusion. The inverse transform is evaluated at eightfold correlation sampling. Only candidate lags within $\|\mathbf{m}_i - \mathbf{m}_j\|/c$ are searched. A three-point quadratic peak refinement is clipped to one correlation-grid step and then to the physical delay bound. Equation 2 converts the three refined delays to azimuth. These implementation choices define the method evaluated here; “GCC-PHAT” alone would not specify them.

2.5 Acoustic descriptors

Acquired impulse responses produce the reverse-integrated squared response, following the energy-decay approach [7]. We fit T20 over -5 to -25 dB and extrapolate the fitted slope to 60 dB; EDT, T30, and fit quality are retained as supplementary outputs. Energy is summed across microphones before decay fitting. A component direct-to-reverberant ratio is computed from the energies of the separately filtered direct



(b) Paired attribution controls

Arrival construction

Fractional (F) / Rounded (Q)

Source history

Steady (S) / Onset (O)

Propagation

Direct (D) / Reflected (R)

Eight combinations share each source/noise record

Figure 1: Array and paired controls. The source-direction arrow indicates azimuth; its length is schematic, while the microphone coordinates are to scale. The source distance is 1.5 m. Each of the eight combinations shares the same underlying source and microphone-noise samples for a given scene and record. Onset means one startup at the beginning of the complete recording.

and reflected responses. Their coherent cross term is also retained. This ratio does not presume that direct and reflected energy add without interference. Direct and all six first-order reflection times and gains accompany each scene.

3 Verification and experimental design

3.1 Independent controls and numerical tolerance

An independent scalar implementation solves the two-by-two normal equations without importing the production geometry solve. Plane-wave recovery, spherical propagation, cardinal directions, and all six channel permutations are checked. The analytical sweep spans 720 angles, radii 0.025/0.05/0.10 m, source distances 0.5/1.5/5/20 m, and rates 16/48/96/192 kHz; 34,560 conditions. A separate harmonic oracle compares directly evaluated continuous-time signals with fractional-delay FIR signals at matched physical bandwidth, duration, and observation time.

The room reference enumerates reflected tiles through a different integer/parity construction and evaluates each path with its exact frequency-domain phase. It imports neither the production enumeration nor the fractional response builder. Tests compare path counts, spectra, and resulting angle estimates, including unequal wall coefficients. Image tent is refined until two successive changes are below 0.05° in angle, 1% in T20, and 0.1% in probe spectrum. Response duration is checked separately at fixed extent. Oversampling 8, 16, and 32 is compared with the independent reference; the production factor 16 and the finer factor 32 must pass the declared spectral and angular tolerances.

The 0.05° discrepancy target is chosen for interpreting effects of roughly 0.5° or larger. It is not an acoustic

Table 1: Frozen confirmatory allocation. A frame is 2048 samples. Counts exclude exploratory and numerical-reference work.

Study	Scenes	Evaluations
Core: 3 SNRs, 5×32 frames	216	103,680
Attribution: 7 additional variants	72	80,640
Long records: 10×128 frames	72	92,160
Primary total		276,480

tic accuracy requirement. Refinement on stochastic recordings is also necessary because a small response change could alter selection between nearly tied peaks. Exploratory pilot inputs and confirmatory inputs have separate identity-derived random streams. Numerical exceptions are retained rather than silently removed.

3.2 Frozen campaign

Table 1 summarizes the protocol, frozen before confirmatory results were generated. The rooms are $6 \times 5 \times 3$, $5 \times 4 \times 2.8$, and $8 \times 6 \times 3.2$ m. Array centers are at room-coordinate fractions (0.5, 0.5, 0.5) and (0.6, 0.45, 0.5). Nominal decay settings are 0.15, 0.30, and 0.60 s. Directions are $7.37^\circ + 30^\circ j$, $j = 0, \dots, 11$. All coordinates are inside the rooms, with minimum source/microphone wall clearance 0.312 m.

The paired and long-record subset uses six acoustic states: the centered, nominal-0.15-s state and the offset, nominal-0.60-s state in each room. Including 12 directions gives 72 physical scenes. We call these the low and high subsets, respectively. Placement and absorption differ between them; their difference is not a controlled single-factor RT comparison. The core matrix, in contrast, includes both placements at every nominal decay setting.

The eight attribution variants cross fractional/rounded arrivals, steady/onset history, and direct/reflected propagation (Fig. 1). At 10 dB, five 32-frame records are shared across all eight conditions.

The corrected fractional/steady/reflected condition is reused exactly from the core matrix. In each of the 72 long-record scenes, five new 128-frame recordings estimate bias/covariance and five different recordings test averaging. These streams do not overlap with pilot or core records.

The fixed secondary subset contains 24 scenes: four directions in each of the six states. Five 32-frame records at 10 dB assess a 17° array rotation, a non-equilateral triangle, unequal wall absorption, and a 300–1700 Hz source band. The estimator and acquisition band remain 300–3400 Hz in the latter test. Variations use separate random namespaces and are scope probes rather than paired causal contrasts with the original geometry. Baseline SRP-PHAT uses the same input frames, band, window, and regularization, but sums pair cross spectra at continuous predicted plane-wave delays over azimuth. A 0.25° angular grid with quadratic refinement is checked against a 0.05° grid. Reverberant localization has a long-established algorithmic literature [8]; this is a limited matched comparison, not an estimator ranking across applications.

3.3 Errors and uncertainty

Signed error is wrapped to $[-180^\circ, 180^\circ)$. We report RMSE, median and 90th-percentile absolute error, circular bias, and the proportion exceeding 5° . The latter is a predeclared descriptive threshold, not a perceptual or device-acceptance standard. Error tails and difficult scenes remain in every aggregate.

We use 2000 bootstrap draws [9], resampling whole recordings with replacement within each fixed scene. Paired variants use the same selected records. Thus the intervals describe source/noise variability conditional on the selected geometries and directions. Three room geometries do not support a population-of-rooms uncertainty statement. Individual scene summaries accompany balanced pooled estimates, and the 160 frames in a short-record scene are not counted as 160 independent experimental replicates.

Attribution contrasts use paired signed errors and paired squared errors. We do not subtract RMSE values to manufacture additive error components. Factorial main effects average over the other two factors; interactions use the corresponding differences of differences. Signed wrapped means are descriptive and can conceal multimodal or opposing errors, so squared-error contrasts and the complete distributions remain central.

3.4 Held-out averaging

For $n = 1, 2, 4, 8, 16, 32, 64, 128$, direction estimates are averaged circularly over contiguous nonoverlapping blocks. The longest duration is 5.461 s. A small-error model predicts

$$\text{MSE}(\bar{e}_n) \simeq b^2 + \frac{1}{n} \left[\gamma_0 + 2 \sum_{\ell=1}^{n-1} \left(1 - \frac{\ell}{n} \right) \gamma_\ell \right]. \quad (5)$$

Training records provide circular bias b and record-centered autocovariance estimates. Covariances are

truncated at lag 32 and Bartlett-tapered; negative predicted variances are clipped to zero. A separate independent-frame approximation sets nonzero lags to zero. Predictions receive substantive interpretation only for scenes satisfying the frozen training rule: circular resultant at least 0.95 and absolute circular mean error at most 15° . All scenes retain empirical held-out results. The rule is a concentration screen, not proof that rare errors or nonstationarity are absent.

4 Results

4.1 Verification and realized decay

The exact plane-wave solve agrees to 5.7×10^{-14} degrees, and the independent scalar reference differs by at most 7.3×10^{-12} degrees. At the selected direct-path FIR half-width 32, the largest oracle/FIR angular discrepancy is 3.7×10^{-5} degrees. Increasing GCC correlation sampling from 8 to 64 changes direct-path estimates by at most 0.000495° .

At full selected image coverage, the most demanding independent room check compares approximately 2.15 million paths per microphone. Relative spectral discrepancy is 3.20×10^{-5} and angle discrepancy is 0.00181° . Coarse image coverage changed one high-setting estimate by more than 100° before convergence. A coarse fractional grid also missed the spectral target; both findings are documented in the numerical notes. They motivated refinement, not relaxation of the tolerance.

On the confirmatory secondary subset, increasing propagation oversampling from 16 to 32 changes individual estimates by at most 0.02048° ; image extent 48 to 56 changes them by 0.00424° ; GCC factor 8 to 16 changes them by 0.01399° . SRP grid refinement changes estimates by at most 0.00305° . No checked stochastic frame exceeds 0.05° (Fig. 2). These selected checks do not constitute a worst-case theorem for every possible stochastic input.

Realized band T20 ranges are 0.083–0.110 s, 0.259–0.295 s, and 0.586–0.683 s for nominal settings 0.15, 0.30, and 0.60 s, respectively. Minimum T20 fit R^2 is 0.928 in the low setting and 0.994 in the other settings. A nominal RT label therefore cannot replace response measurement. Component DRR spans approximately +9.53 to -10.77 dB across the full matrix.

4.2 Corrected room performance

At 10 dB, the corrected core RMSE is 1.616° [95% interval 1.597, 1.635] for nominal 0.15 s, 12.112° [11.253, 12.956] for 0.30 s, and 37.901° [37.078, 38.759] for 0.60 s. The proportions above 5° are 0.77%, 27.40%, and 62.03%. The corresponding 20-dB RMSEs are 1.329, 11.495, and 37.328° : reducing additive noise does not eliminate reflected-field errors in this model.

Figure 3 preserves differences across rooms. At 10 dB, individual scene RMSE ranges from 0.84 to 2.86° in the lowest setting, 2.03 to 47.53° in the middle setting, and 4.47 to 83.48° in the highest. These ranges are much wider than uncertainty intervals on pooled

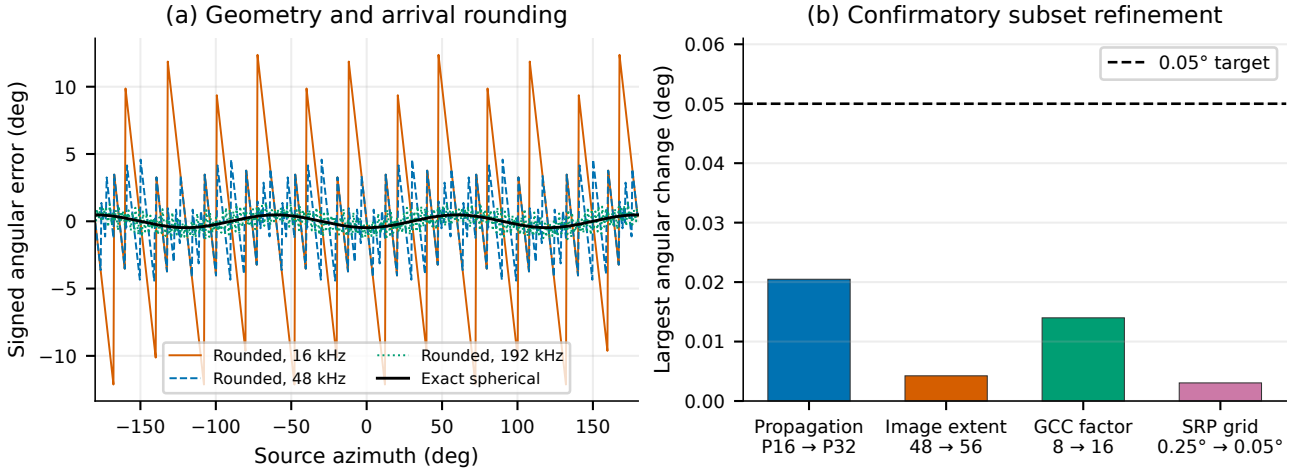


Figure 2: Numerical effects and checks. (a) Analytical arrival rounding at $R = 0.05$ m and source distance 1.5 m; exact spherical arrivals are interpreted by the same far-field solve. These are not noisy waveform-estimation results. (b) Maximum individual angular changes over the fixed 24-scene confirmatory subset, five records and 32 frames per scene. The dashed line is a numerical-discrepancy target.

conditional means. Direction, placement, and room geometry are therefore material parts of the reported result.

4.3 Paired rounding and source-history effects

Across the 72 attribution scenes at 10 dB, steady direct-path RMSE is 0.929° [0.920, 0.939] with fractional arrivals and 2.279° [2.266, 2.293] with rounding. The paired increase in mean squared error is 4.330 deg^2 [4.272, 4.388]. This establishes a rounding effect after waveform processing, separately from the analytical geometry diagnostic.

With reflections, steady RMSE is 28.389° for fractional arrivals and 28.729° after rounding. The paired squared-error change is 19.409 deg^2 , but its interval $[-2.396, 42.332]$ spans zero. Thus, the clear direct-path rounding penalty does not justify a universal same-sign penalty in a coherent reflected field. Rounding changes interference and can alter which peak is selected.

The source-history contrast is strongest immediately after startup (Fig. 4). In the high subset, the first-frame RMSE is 4.591° at onset versus 40.415° with populated prehistory. By the last frame of the 1.365-s short record, both reach 38.746° for the matched realizations. For fractional reflected signals pooled across all 72 scenes and 32 frames, onset reduces mean squared error by 56.011 deg^2 [36.349, 77.029], expressed as a reduction; onset-minus-steady is negative. This describes transient observation, not an improvement to the estimator.

The factorial onset-by-reflection interaction is -52.601 deg^2 $[-70.892, -33.762]$. The corresponding rounding-by-reflection interval spans zero. Interactions and broad error tails make an independent additive error budget inappropriate. The separate exploratory per-frame-reset diagnostic gives much lower reflected RMSE than continuous records, but it uses pilot directions and is not substituted for the confirmatory time

trajectory.

4.4 Independent-record averaging

Held-out results confirm a reduction in dispersion with longer averaging, without a common residual accuracy across scenes (Fig. 5). In the low subset, RMSE changes from 1.598° [1.585, 1.612] for one frame to 0.842° [0.824, 0.861] for 128 frames. In the high subset, it changes from 39.755° [39.117, 40.415] to 10.041° [9.298, 10.700]. At 128 frames, 45% of high-subset block estimates still exceed 5° .

Forty-eight of 72 scenes pass the training concentration rule. Their held-out 128-frame RMSE is 2.034° [1.934, 2.133], compared with 2.002° from the bias/covariance approximation and 2.028° from the independent-frame approximation. This close aggregate agreement is compatible with the model on the selected set; the tiny difference between the two approximations does not establish a useful superiority of covariance correction. The empirical results for the excluded, less concentrated scenes remain in panel (a) and the supplement. A 5.46-s observation is finite: it does not establish an asymptotic or universal reverberation floor.

4.5 Precision criterion and scope probes

Let B be the pair-incidence matrix, so $A = BM$. Perturbing microphone arrivals by ϵ changes the solved direction vector exactly by

$$\delta \tilde{\mathbf{u}} = J\epsilon, \quad J = -cA^+B. \quad (6)$$

This retains dependence between pairs sharing a microphone. For $|\epsilon_i| \leq h$, compute $\rho = \max_{\epsilon_i \in \{-h, h\}} \|J\epsilon\|$. If $\rho < \|\tilde{\mathbf{u}}\|$, the exact angular extreme is attained at a vertex of the transformed box, and a simpler conservative bound is

$$|\delta\theta| \leq \arcsin \frac{\rho}{\|\tilde{\mathbf{u}}\|}. \quad (7)$$

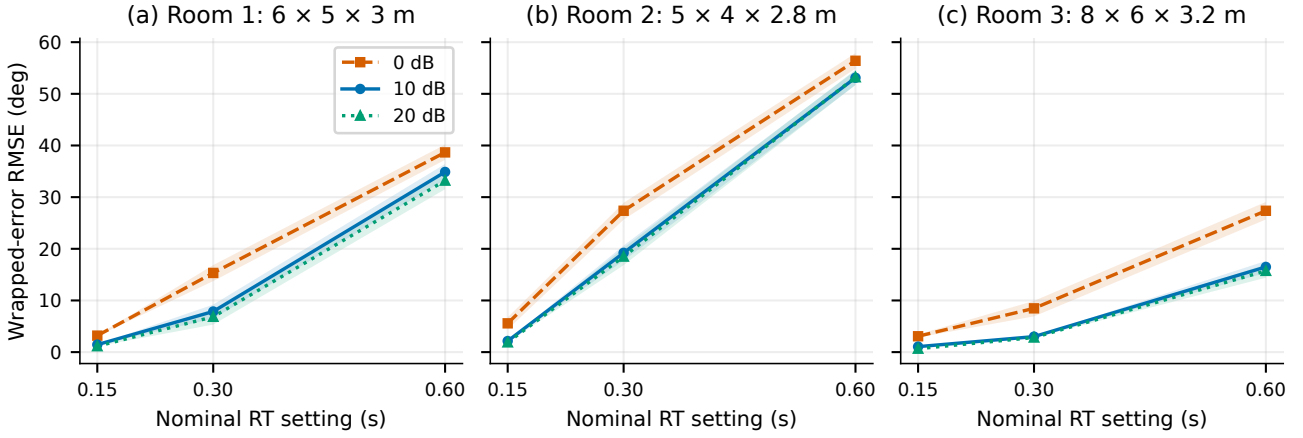


Figure 3: Corrected fractional/steady/reflected core results, with rooms shown separately. Each point pools two placements, 12 directions and five 32-frame records. Bands are 95% bootstrap intervals conditional on those fixed scenes. The horizontal coordinate is the nominal absorption/RT setting, not an assertion of realized RT.

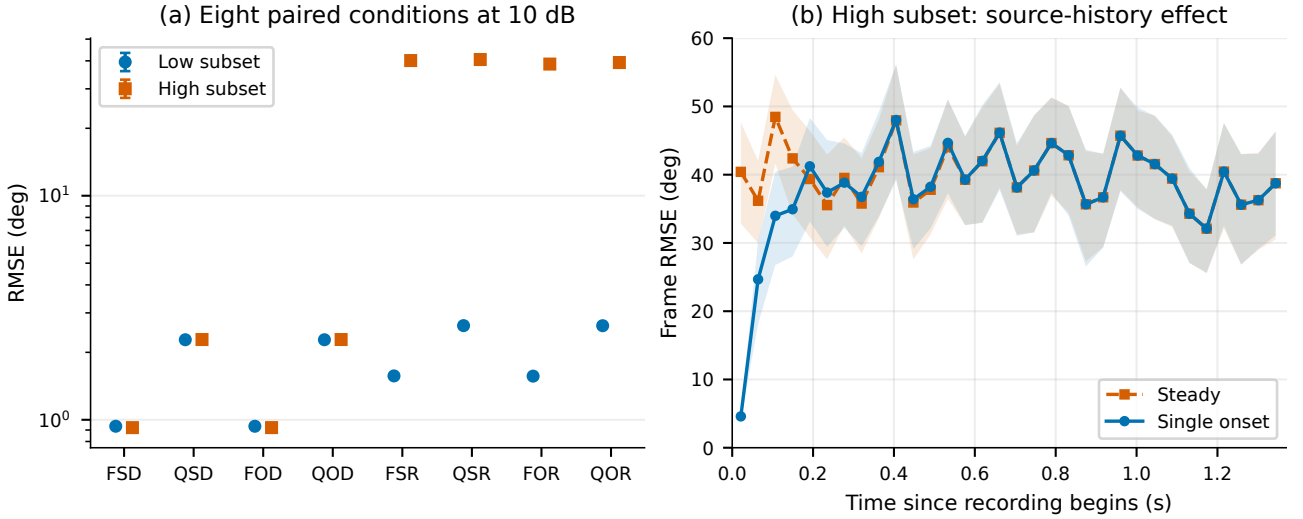


Figure 4: Paired attribution at 10 dB. (a) Eight combinations: F/Q, fractional/rounded; S/O, steady/onset; D/R, direct/reflected. Low and high subsets each contain 36 scenes; their placement and absorption settings differ. Error bars are record-bootstrap intervals. (b) Matched fractional reflected recordings in the high subset. The first onset frame contains little established reflected energy; later frames approach the steady condition. Time is measured from the beginning of the analyzed recording, and markers show frame midpoints.

For a centered equilateral array and a plane wave, $\rho = 4ch/(3R)$ and $\|\tilde{\mathbf{u}}\| = 1$. A sufficient condition for perturbation below η is $h \leq (3R/4c) \sin \eta$. At $R = 0.05$ m and $\eta = 0.5^\circ$, this permits at most 0.954 microseconds of absolute error per arrival. It is not a necessary sampling rate or a guarantee for GCC peak selection. All 34,560 rounded-arrival cases satisfy the exact vertex bound; 12,000 additional random perturbations, including unequal triangles and an independent scalar solve, show no violations. Figure 6 illustrates the bound on change relative to exact spherical estimates. Finite-distance mismatch relative to true source azimuth remains a separate effect.

The secondary subset demonstrates the limits of extrapolation. GCC-PHAT RMSE is 22.682° in its base condition, 23.609° after rotation, 18.916° with the equal triangle, 21.114° with unequal walls, and 42.496° for the lower-band source. These variations use new

random records and do not define a geometry optimization. The lower-band test also changes how much of the fixed estimator band contains useful source energy.

On the same base frames, SRP-PHAT has 29.909° RMSE versus GCC-PHAT's 22.682° , but a lower 5° exceedance rate, 28.70% versus 31.09%. The paired squared-error difference is 380.063 deg^2 [290.101, 469.786]. The disagreement between RMSE and threshold rate reflects tail behavior; neither scalar alone supports a universal ranking. The complete method and scene summaries are retained.

5 Discussion

The central practical result is that numerical and recording choices can alter the acoustic question being answered. Rounding continuous path arrivals creates an observable direct-path penalty. Starting with an

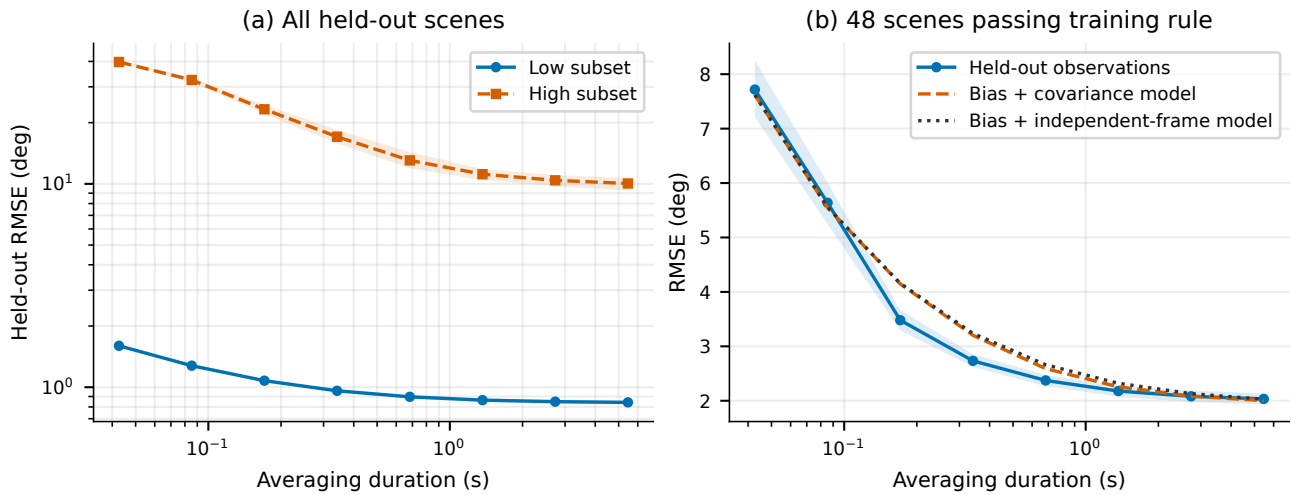


Figure 5: Averaging on independent held-out recordings. (a) Circular block means in all 72 scenes, separated into low and high subsets. (b) Only the 48 scenes passing the rule determined from separate training records. Predictions are model outputs, not confidence intervals. Shaded intervals resample whole held-out recordings within each fixed scene.

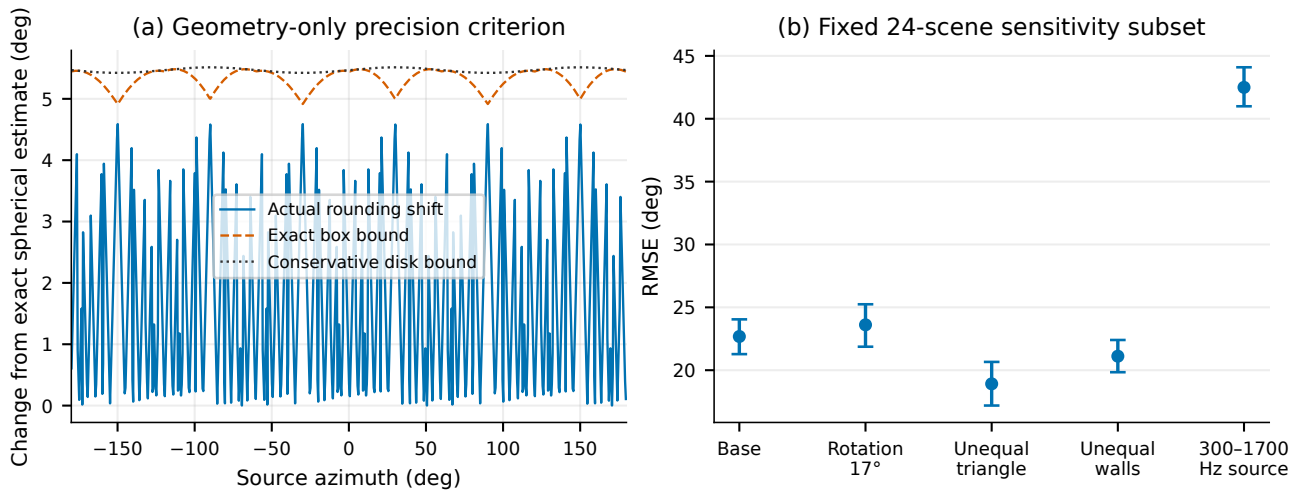


Figure 6: Precision and limited transfer checks. (a) Actual analytical rounding shift and deterministic bounds at 48 kHz, radius 0.05 m and source distance 1.5 m. The ordinate is relative to exact spherical-arrival estimates, not acoustic truth. (b) Five fixed sensitivity conditions; points are conditional RMSEs with record-bootstrap intervals. The source-band change retains a 300–3400 Hz estimator band.

empty source history produces an even larger appar-
 ent gain in an early reflected-field frame. A simulation
 that restarts the source for every short frame repeat-
 edly samples this transient regime and cannot support
 a claim about continuous steady operation.

The appropriate control is paired propagation and
 source history at a fixed direct-referenced noise level,
 with independent numerical checks preceding interpre-
 tation. The direct-path and reflected-field rounding
 contrasts need not have the same sign: coherent in-
 terference and peak selection prevent simple addition
 of separate error magnitudes. Image coverage also mat-
 ters; a nominal response duration without a sufficient
 image domain can omit influential paths. Reporting an
 achieved discrepancy at selected checks is more infor-
 mative than only stating a large oversampling or image
 count.

The residual under averaging is conditional on the es-
 timator and selected scenes. Longer averaging reduces

random fluctuation but does not change the room trans-
 fer functions, finite-distance mismatch, or a stable pref-
 erence for an incorrect correlation peak. Conversely,
 a finite residual at the longest tested duration is not
 proof of an infinite-time floor. The training/held-out
 split limits fitting to the same records used for evalua-
 tion; it does not convert three prescribed rooms into a
 representative population.

Several limitations constrain transfer. Ideal matched
 omnidirectional microphones avoid calibration, clock
 drift, gain mismatch, and platform artifacts. The
 source is synthetic band-limited Gaussian noise rather
 than speech or an operating machine. The room model
 is coherent, specular, and frequency-independent; real
 rooms may have different decay, scattering, and di-
 rectional structure. The main source distance and ar-
 ray size are fixed, despite the broader analytical sweep.
 Rare peak switches can remain difficult to predict, and
 the numerical tolerance has been verified on specified

controls and a fixed stochastic subset, not every possible signal.

Bootstrap intervals use only five independent records per short-record scene. They are useful conditional summaries, but their coverage has not been independently calibrated and the individual-scene intervals have limited empirical support. The balanced matrix and per-scene outputs make this limitation visible. The most useful release is therefore the complete computational record: configurations, input identities, exact method definitions, difficult outcomes, and reproducible figures. An external measured-room study would be an extension requiring new validation evidence, not a relabeling of the present numerical checks.

6 Conclusion

For the specified three-microphone model, arrival rounding and omitted source history produce distinct changes in estimated azimuth. Independent references and refinement keep the checked numerical discrepancy well below the principal interpreted effects. The matched startup experiment shows how a large apparent accuracy improvement can result from observing an unestablished reflected field. Held-out averaging reduces error but retains strong scene dependence. The geometry-only timing bound supplies a conservative analytical precision check with explicit limits; it does not certify noisy waveform localization. These results support careful simulation attribution and reproducibility rather than a universal acoustic error law.

Declarations and availability

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Conflicts of interest. Author confirmation is pending for this review version. No absence-of-conflict declaration is inferred.

Contributions and assistance. A.A.C. selected the research direction and funded the work. Final scientific review and release approval remain pending. OpenAI Codex assisted substantively with planning, derivations, implementation, reference checks, literature synthesis, statistical analysis, figures, and manuscript drafting. The figures are deterministic plots generated with AI-authored code from retained results. The reference implementations and checks are internal computational verification; they are not external peer review. The assistance was not limited to language editing.

Data and software. The accompanying server-neutral reproducibility bundle contains per-frame results, configurations, source code, figure sources, environment records, verification reports, and reproduction instructions. No public repository or archive identifier has yet been assigned. Distribution licenses and public release await author decision. The supplement gives additional mathematical, statistical, and reproduction details. The local review package makes no claim of journal acceptance or editorial-policy eligibility.

References

- [1] C. Knapp and G. Carter. The generalized correlation method for estimation of time delay. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 24(4):320–327, 1976. doi: 10.1109/TASSP.1976.1162830.
- [2] Hilman Adritya H.B.B. Catur and Hendri Maja Saputra. Azimuth Estimation based on Generalized Cross Correlation Phase Transform (GCC-PHAT) Using Equilateral Triangle Microphone Array. In *2019 International Conference on Radar, Antenna, Microwave, Electronics, and Telecommunications (ICRAMET)*, pages 89–93. IEEE, 2019. doi: 10.1109/ICRAMET47453.2019.8980432.
- [3] T.I. Laakso, V. Valimaki, M. Karjalainen, and U.K. Laine. Splitting the unit delay [FIR/all pass filters design]. *IEEE Signal Processing Magazine*, 13(1): 30–60, 1996. doi: 10.1109/79.482137.
- [4] Jont B. Allen and David A. Berkley. Image method for efficiently simulating small-room acoustics. *The Journal of the Acoustical Society of America*, 65(4): 943–950, 1979. doi: 10.1121/1.382599.
- [5] Augustin Ernoult, Romain Viala, Jérémy Cabaret, Juliette Chabassier, Tom Colinot, Jean-Pierre Dalmont, Jean-Baptiste Doc, and Vincent Fréour. Benchmark study of pipe input impedance simulations and measurements for verification and validation in musical acoustics context. *Acta Acustica*, 10:51, 2026. doi: 10.1051/aacus/2026048.
- [6] Samuel D. Bellows and Brian F. G. Katz. Constrained least-squares and maximum-likelihood calibration of absorption coefficients in reverberation time equations. *Acta Acustica*, 9:25, 2025. doi: 10.1051/aacus/2025011.
- [7] M. R. Schroeder. New Method of Measuring Reverberation Time. *The Journal of the Acoustical Society of America*, 37(3):409–412, 1965. doi: 10.1121/1.1909343.
- [8] Joseph H. DiBiase, Harvey F. Silverman, and Michael S. Brandstein. Robust Localization in Reverberant Rooms. In *Microphone Arrays: Signal Processing Techniques and Applications*, pages 157–180. Springer, 2001. doi: 10.1007/978-3-662-04619-7_8.
- [9] B. Efron. Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.